

TEXT CLASSIFICATION

```
!pip uninstall -y transformers datasets huggingface_hub tokenizers accelerate
```

```
Found existing installation: transformers 5.12.1
Uninstalling transformers-5.12.1:
  Successfully uninstalled transformers-5.12.1
Found existing installation: datasets 4.0.0
Uninstalling datasets-4.0.0:
  Successfully uninstalled datasets-4.0.0
Found existing installation: huggingface_hub 1.20.1
Uninstalling huggingface_hub-1.20.1:
  Successfully uninstalled huggingface_hub-1.20.1
Found existing installation: tokenizers 0.22.2
Uninstalling tokenizers-0.22.2:
  Successfully uninstalled tokenizers-0.22.2
Found existing installation: accelerate 1.14.0
Uninstalling accelerate-1.14.0:
  Successfully uninstalled accelerate-1.14.0
```

```
!pip install \
transformers==4.46.3 \
datasets==3.2.0 \
huggingface_hub==0.26.2 \
tokenizers==0.20.3 \
accelerate==1.1.1
```

```
Requirement already satisfied: transformers==4.46.3 in /usr/local/lib/python3.12/dist-packages (4.46.3)
Requirement already satisfied: datasets==3.2.0 in /usr/local/lib/python3.12/dist-packages (3.2.0)
Requirement already satisfied: huggingface_hub==0.26.2 in /usr/local/lib/python3.12/dist-packages (0.26.2)
Requirement already satisfied: tokenizers==0.20.3 in /usr/local/lib/python3.12/dist-packages (0.20.3)
Requirement already satisfied: accelerate==1.1.1 in /usr/local/lib/python3.12/dist-packages (1.1.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (3.29.7)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (26.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (2024.11.12)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (2.32.4)
Requirement already satisfied: safetensors>=0.4.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (0.5.2)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (4.67.3)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets==3.2.0) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets==3.2.0) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets==3.2.0) (2.2.2)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets==3.2.0) (3.8.1)
Requirement already satisfied: multiprocessing<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets==3.2.0) (0.70.17)
Requirement already satisfied: fsspec<=2024.9.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from datasets==3.2.0) (2024.9.0)
Requirement already satisfied: aiohttp in /usr/local/lib/python3.12/dist-packages (from datasets==3.2.0) (3.14.1)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub==0.26.2) (4.12.2)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate==1.1.1) (5.9.5)
Requirement already satisfied: torch>=1.10.0 in /usr/local/lib/python3.12/dist-packages (from accelerate==1.1.1) (2.11.0+cpu)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp->datasets==3.2.0) (2.5.0)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp->datasets==3.2.0) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp->datasets==3.2.0) (26.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp->datasets==3.2.0) (1.5.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp->datasets==3.2.0) (6.1.0)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp->datasets==3.2.0) (0.2.0)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp->datasets==3.2.0) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.46.3) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.46.3) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.46.3) (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.46.3) (2025.1.1)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1) (75.1.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1) (1.13.3)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1) (3.4.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1) (3.1.4)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets==3.2.0) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets==3.2.0) (2025.1)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets==3.2.0) (2025.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->datasets==3.2.0) (1.17.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.10.0->accelerate==1.1.1) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.10.0->accelerate==1.1.1) (3.0.2)
```

```
import transformers
import datasets
import huggingface_hub

print(transformers.__version__)
print(datasets.__version__)
print(huggingface_hub.__version__)
```

```
4.46.3
3.2.0
```

0.26.2

Using a Task-Specific Model

```
# Imports the load_dataset() function from the Hugging Face datasets library.
from datasets import load_dataset

# Downloads and loads the Rotten Tomatoes sentiment dataset into the variable data.
data = load_dataset("rotten_tomatoes")
print(data)
```

```
DatasetDict({
  train: Dataset({
    features: ['text', 'label'],
    num_rows: 8530
  })
  validation: Dataset({
    features: ['text', 'label'],
    num_rows: 1066
  })
  test: Dataset({
    features: ['text', 'label'],
    num_rows: 1066
  })
})
```

```
# retrieves the first and last examples from the training dataset.
data["train"][0, -1]
```

```
{'text': ['the rock is destined to be the 21st century\'s new " conan " and that he\'s going to make a splash even greater
than arnold schwarzenegger , jean-claud van damme or steven segal .'],
 'label': [1, 0]}
```

```
# from transformers library importing pipeline
from transformers import pipeline

# Name (path) of the Hugging Face model to use
model_path = "cardiffnlp/twitter-roberta-base-sentiment-latest"

# Load model into pipeline
pipe = pipeline(
    # Load this pre-trained model
    model=model_path,
    # Load the matching tokenizer for the model
    tokenizer=model_path,

    # return the prediction score for every possible class
    return_all_scores=True,

    # Run the model on the CPU
    device="cpu"
)

# Predict the sentiment of the given sentence
pipe("I hate this movie")
```

```
config.json: 100% 929/929 [00:00<00:00, 77.0kB/s]
```

```
pytorch_model.bin: 100% 501M/501M [00:06<00:00, 72.0MB/s]
```

Some weights of the model checkpoint at cardiffnlp/twitter-roberta-base-sentiment-latest were not used when initializing RobertaForSequenceClassification from the checkpoint of a model trained on another task (e.g. sequence classification vs. token classification). This is expected if you are initializing RobertaForSequenceClassification from the checkpoint of a model trained on another task (e.g. sequence classification vs. token classification). This is NOT expected if you are initializing RobertaForSequenceClassification from the checkpoint of a model that you expect to be initialized with the weights for the task you are initializing it for.

```
vocab.json: 899k/? [00:00<00:00, 34.6MB/s]
```

```
merges.txt: 456k/? [00:00<00:00, 30.0MB/s]
```

```
special_tokens_map.json: 100% 239/239 [00:00<00:00, 27.5kB/s]
```

```
/usr/local/lib/python3.12/dist-packages/transformers/pipelines/text_classification.py:106: UserWarning: `return_all_scores`
warnings.warn(
```

```
[[{'label': 'negative', 'score': 0.8983777165412903},
 {'label': 'neutral', 'score': 0.08257268369197845},
 {'label': 'positive', 'score': 0.019049596041440964}]]
```

```
import numpy as np
```

```
# tqdm = It creates a progress bar
from tqdm import tqdm
```

```
# Import KeyDataset, which lets the pipeline read only the review text from the dataset.
from transformers.pipelines.pt_utils import KeyDataset

# Create an empty list that will store all the model's predictions.
y_pred = []

# 1 Select only the review column from the test data set
# 2 run the sentiment model
# 3 show a progress bar
# 4 store each prediction in the variable output.
for output in tqdm(pipe(KeyDataset(data["test"], "text")),

                    # Len of the test dataset
                    total=len(data["test"])):

    # Model predicts the output as label and score .It predicts three labels for each review
    # The index for
    # 0 = negative
    # 1 = neutral
    # 2 = postive
    # Store the negative sentiment score predicted by the model.
    negative_score = output[0]["score"]

    # Store the postive sentiment score predicted by the model
    positive_score = output[2]["score"]

    # Selecting the max score among those two
    assignment = np.argmax([negative_score, positive_score])

    # prediction is added to list
    y_pred.append(assignment)
```

100%|██████████| 1066/1066 [02:42<00:00, 6.57it/s]

```
# Imports the classification_report function to evaluate the classification model.
from sklearn.metrics import classification_report

# Creates a function that compares the actual labels with the predicted labels.
def evaluate_performance(y_true, y_pred):

    # classification_report() = Compares the actual labels (y_true) with the predicted labels (y_pred).
    performance = classification_report(
        y_true, y_pred,

        # Replaces numeric labels with readable names.
        target_names=["Negative Review", "Positive Review"]
    )
    print(performance)

# Inputs are revies from test data and y_predicted
evaluate_performance(data["test"]["label"], y_pred)
```

	precision	recall	f1-score	support
Negative Review	0.76	0.88	0.81	533
Positive Review	0.86	0.72	0.78	533
accuracy			0.80	1066
macro avg	0.81	0.80	0.80	1066
weighted avg	0.81	0.80	0.80	1066

Classification Tasks That Leverage Embeddings

```
# It deletes the cached copy of the Hugging Face model sentence-transformers/all-mpnet-base-v2 from your local machine or (
!rm -rf ~/.cache/huggingface/hub/models--sentence-transformers--all-mpnet-base-v2
```

```
# It installs or upgrades three Python libraries to their latest versions.
!pip install -U sentence-transformers transformers accelerate
```

```
Requirement already satisfied: sentence-transformers in /usr/local/lib/python3.12/dist-packages (5.6.0)
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.14.0)
Requirement already satisfied: accelerate in /usr/local/lib/python3.12/dist-packages (1.14.0)
Requirement already satisfied: huggingface-hub>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (0.23.0)
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (2.11.0)
Requirement already satisfied: numpy>=1.20.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (2.0.2)
Requirement already satisfied: scikit-learn>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.3.2)
Requirement already satisfied: scipy>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.16.3)
Requirement already satisfied: typing_extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (4.6.3)
Requirement already satisfied: tqdm>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (4.67.3)
```

```
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex>=2025.10.22 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.20.3)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from transformers) (0.25.1)
Requirement already satisfied: safetensors>=0.8.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.8.0)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: click<9.0.0,>=8.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0->sentence-transformers) (8.1.8)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0->sentence-transformers) (3.16.1)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0->sentence-transformers) (2025.10.2)
Requirement already satisfied: hf-xet<2.0.0,>=1.5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0->sentence-transformers) (1.2.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.23.0->sentence-transformers) (0.27.2)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.22.0->sentence-transformers) (1.4.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.22.0->sentence-transformers) (3.5.0)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (75.8.1)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (1.13.3)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (3.4.1)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (3.1.4)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (0.0.2)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.23.0->sentence-transformers) (4.6.2)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.23.0->sentence-transformers) (2025.11.12)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.23.0->sentence-transformers) (1.0.7)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.23.0->sentence-transformers) (3.10)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub>=0.23.0->sentence-transformers) (0.14.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer->transformers) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer->transformers) (2.19.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.11.0->sentence-transformers) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch>=1.11.0->sentence-transformers) (3.0.2)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer->transformers) (0.1.2)
```

```
# Import the class used to load pretrained embedding models
from sentence_transformers import SentenceTransformer

# Downloads the pretrained transformer that helps to convert into embeddings
model = SentenceTransformer("sentence-transformers/all-mpnet-base-v2")
```

```
HTTP Error 504 thrown while requesting HEAD https://huggingface.co/sentence-transformers/all-mpnet-base-v2/resolve/main/.gitattributes
WARNING:huggingface_hub.utils._http:HTTP Error 504 thrown while requesting HEAD https://huggingface.co/sentence-transformers/all-mpnet-base-v2/resolve/main/.gitattributes
Retrying in 1s [Retry 1/5].
WARNING:huggingface_hub.utils._http:Retrying in 1s [Retry 1/5].
HTTP Error 504 thrown while requesting HEAD https://huggingface.co/sentence-transformers/all-mpnet-base-v2/resolve/main/.gitattributes
WARNING:huggingface_hub.utils._http:HTTP Error 504 thrown while requesting HEAD https://huggingface.co/sentence-transformers/all-mpnet-base-v2/resolve/main/.gitattributes
Retrying in 2s [Retry 2/5].
WARNING:huggingface_hub.utils._http:Retrying in 2s [Retry 2/5].
HTTP Error 504 thrown while requesting HEAD https://huggingface.co/sentence-transformers/all-mpnet-base-v2/resolve/main/.gitattributes
WARNING:huggingface_hub.utils._http:HTTP Error 504 thrown while requesting HEAD https://huggingface.co/sentence-transformers/all-mpnet-base-v2/resolve/main/.gitattributes
Retrying in 4s [Retry 3/5].
WARNING:huggingface_hub.utils._http:Retrying in 4s [Retry 3/5].
HTTP Error 504 thrown while requesting HEAD https://huggingface.co/sentence-transformers/all-mpnet-base-v2/resolve/main/.gitattributes
WARNING:huggingface_hub.utils._http:HTTP Error 504 thrown while requesting HEAD https://huggingface.co/sentence-transformers/all-mpnet-base-v2/resolve/main/.gitattributes
Retrying in 8s [Retry 4/5].
WARNING:huggingface_hub.utils._http:Retrying in 8s [Retry 4/5].
HTTP Error 504 thrown while requesting HEAD https://huggingface.co/sentence-transformers/all-mpnet-base-v2/resolve/main/.gitattributes
WARNING:huggingface_hub.utils._http:HTTP Error 504 thrown while requesting HEAD https://huggingface.co/sentence-transformers/all-mpnet-base-v2/resolve/main/.gitattributes
Retrying in 8s [Retry 5/5].
WARNING:huggingface_hub.utils._http:Retrying in 8s [Retry 5/5].
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast downloads.

modules.json: 100% 349/349 [00:00<00:00, 5.30kB/s]

config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 3.11kB/s]

README.md: 100% 11.6k/11.6k [00:00<00:00, 115kB/s]

sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 847B/s]

config.json: 100% 571/571 [00:00<00:00, 8.96kB/s]

model.safetensors: reconstructing file: 100% 438MB / 438MB, 39.3MB/s

model.safetensors: downloading bytes: 414MB, 36.1MB/s

Loading weights: 100% 199/199 [00:00<00:00, 3816.16it/s]

tokenizer_config.json: 100% 363/363 [00:00<00:00, 31.5kB/s]

vocab.txt: 100% 232k/232k [00:00<00:00, 11.8MB/s]

tokenizer.json: 100% 466k/466k [00:00<00:00, 21.7MB/s]

special_tokens_map.json: 100% 239/239 [00:00<00:00, 16.8kB/s]

config.json: 100% 190/190 [00:00<00:00, 13.7kB/s]
```

```
# model = Embedding model.
# .encode() = Converts text into embeddings.
```

```
# data["train"]["text"] = All training reviews.
# show_progress_bar=True Show a progress bar while encoding.
train_embeddings = model.encode(data["train"]["text"],
show_progress_bar=True)

# Same for test data
test_embeddings = model.encode(data["test"]["text"],
show_progress_bar=True)
```

Batches: 100% 267/267 [16:13<00:00, 1.01it/s]
 Batches: 100% 34/34 [02:03<00:00, 1.30s/it]

```
print(train_embeddings.shape)
```

```
(8530, 768)
```

```
# Imports the Logistic Regression classifier.
from sklearn.linear_model import LogisticRegression

#clf → Variable name (classifier).
# random_state=42 = If you run the code again, you should get the same outcome.
clf = LogisticRegression(random_state=42)

# train the model with training embeddings which are review data in train dataset
clf.fit(train_embeddings, data["train"]["label"])
```

```
LogisticRegression
LogisticRegression(random_state=42)
```

```
# Import the function used to evaluate classification models
from sklearn.metrics import classification_report

# # Use the trained classifier to predict the labels of the test data
y_pred = clf.predict(test_embeddings)

# Function to evaluate the model's performance
def evaluate_performance(y_true, y_pred):
    """Create and print the classification report"""
    performance = classification_report(
        y_true, y_pred,
        target_names=["Negative Review", "Positive Review"]
    )
    print(performance)

    evaluate_performance(data["test"]["label"], y_pred)
```

	precision	recall	f1-score	support
Negative Review	0.85	0.86	0.85	533
Positive Review	0.86	0.85	0.85	533
accuracy			0.85	1066
macro avg	0.85	0.85	0.85	1066
weighted avg	0.85	0.85	0.85	1066

```
# WITHOUT LABELLED DATA
```

```
# Create embeddings for our labels
label_embeddings = model.encode(["A negative review", "A positive review"])
```

```
import numpy as np

# Imports the function that compares two embeddings.
from sklearn.metrics.pairwise import cosine_similarity

# Compare every review with every label
sim_matrix = cosine_similarity(test_embeddings, label_embeddings)

# This chooses the largest similarity score in each row.
y_pred = np.argmax(sim_matrix, axis=1)

# Function call it prints the report
evaluate_performance(data["test"]["label"], y_pred)
```

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

Negative Review	0.78	0.77	0.78	533
Positive Review	0.77	0.79	0.78	533
accuracy			0.78	1066
macro avg	0.78	0.78	0.78	1066
weighted avg	0.78	0.78	0.78	1066

```
# Using the Text-to-Text Transfer Transformer
```

```
!pip install -U transformers==4.46.3 accelerate
```

```
Requirement already satisfied: transformers==4.46.3 in /usr/local/lib/python3.12/dist-packages (4.46.3)
Requirement already satisfied: accelerate in /usr/local/lib/python3.12/dist-packages (1.14.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (3.29.4)
Requirement already satisfied: huggingface-hub<1.0,>=0.23.2 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (0.23.2)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (26.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (2024.5.15)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (2.32.4)
Requirement already satisfied: tokenizers<0.21,>=0.20 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (0.20.1)
Requirement already satisfied: safetensors>=0.4.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (0.5.2)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (4.67.3)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: torch>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from accelerate) (2.11.0+cpu)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.23.2) (2024.5.15)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.23.2) (4.5.0)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (75.0.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (1.13.3)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.3)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.1.6)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.46.3) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.46.3) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.46.3) (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.46.3) (2024.7.4)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=2.0.0->accelerate) (3.1.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch>=2.0.0->accelerate) (3.0.2)
```

```
# Load our model
from transformers import pipeline

# Creates a pipeline.
pipe = pipeline(
    "text2text-generation",
    model="google/flan-t5-small",
    device="cpu"
)
```

```
WARNING:torchao.kernel.intmm:Warning: Detected no triton, on systems without Triton certain kernels will not work
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
```

```
warnings.warn(
model.safetensors: 100% 308M/308M [00:01<00:00, 171MB/s]

generation_config.json: 100% 147/147 [00:00<00:00, 9.01kB/s]

tokenizer_config.json: 2.54k/? [00:00<00:00, 191kB/s]

spiece.model: 100% 792k/792k [00:00<00:00, 53.7MB/s]

tokenizer.json: 2.42M/? [00:00<00:00, 54.9MB/s]

special_tokens_map.json: 2.20k/? [00:00<00:00, 160kB/s]
```

```
# Creating the prompt
prompt = "Is the following sentence positive or negative? "

# adding every with the prompt and new column is created as t5 it stores prompt along with review
data = data.map(lambda example: {"t5": prompt + example['text']})

print(data)
```

```

Map: 100%                               8530/8530 [00:00<00:00, 13264.20 examples/s]

Map: 100%                               1066/1066 [00:00<00:00, 9264.44 examples/s]

Map: 100%                               1066/1066 [00:00<00:00, 10104.20 examples/s]

DatasetDict({
  train: Dataset({
    features: ['text', 'label', 't5'],
    num_rows: 8530
  })
  validation: Dataset({
    features: ['text', 'label', 't5'],
    num_rows: 1066
  })
  test: Dataset({
    features: ['text', 'label', 't5'],
    num_rows: 1066
  })
})

```

```

# Import tqdm to display a progress bar while processing the data
from tqdm import tqdm

# Import KeyDataset to read one column from the dataset efficiently
from transformers.pipelines.pt_utils import KeyDataset

# Which stores the predicted labels
y_pred = []

# Run the pipeline on each text in the "t5" column of the test dataset
for output in tqdm(pipe(KeyDataset(data["test"], "t5")),
total=len(data["test"])):

    # Get the generated text (either "negative" or "positive")
    text = output[0]["generated_text"]

    # Convert the text label into a numeric label
    # "negative" = 0
    # "positive" = 1
    y_pred.append(0 if text == "negative" else 1)

```

```

0%|          | 0/1066 [00:00<?, ?it/s]/usr/local/lib/python3.12/dist-packages/transformers/generation/utils.py:1375: UserWarning: The `num_beams` argument is deprecated and will be removed in a future version. Please use `num_beam_groups` instead.
100%|██████████| 1066/1066 [02:25<00:00, 7.33it/s]

```

ChatGPT for Classification

USE THIS INSTEAD OF GPT

```

from transformers import pipeline

# Create the FLAN-T5 pipeline
pipe = pipeline(
    "text2text-generation", # Generate text from a text prompt
    model="google/flan-t5-small", # model name
    device=-1 # run on CPU (-1), GPU (0)
)

```

```

# Function to generate a response using the FLAN-T5 model
def chatgpt_generation(prompt, document, model="google/flan-t5-small"):
    """Generate an output based on a prompt and an input document."""

    # Create a conversation with a system message and a user message
    messages = [
        {
            "role": "system",
            "content": "You are a helpful assistant."
        },
        {
            "role": "user",

            # Replace [DOCUMENT] in the prompt with the actual document
            "content": prompt.replace("[DOCUMENT]", document)
        }
    ]

    generation = pipe(
        messages[1]["content"], # Send only the user's prompt to FLAN-T5
        max_new_tokens=50,
    )

```



```

        do_sample=False
    )

    # Return only the generated text
    return generation[0]["generated_text"]

```

```

# Define a prompt template as a base
prompt = """Predict whether the following document is a positive
or negative movie review:
[DOCUMENT]
If it is positive return 1 and if it is negative return 0. Do not
give any other answers.
"""

```

```

# Predict the target using GPT
document = "unpretentious , charming , quirky , original"

chatgpt_generation(prompt, document)

'0'

```

```

# Load the predictions for all test reviews
predictions = [

    # Predict the sentiment for one review
    chatgpt_generation(prompt, doc)

    # Repeat for every review in the test dataset
    for doc in data["test"]["text"]

]

```

```

# Extract predictions
# Convert the model's output into numeric labels
y_pred = [

    # If the model returns "positive", store 1
    1 if pred.lower() == "positive"

    # If the model returns "negative", store 0
    else 0 if pred.lower() == "negative"

    # Otherwise, convert the returned value (0 or 1) to an integer
    else int(pred)

    # Repeat for every prediction
    for pred in predictions

]

# y_pred = [int(pred) for pred in predictions]

```

```

# Evaluate performance
evaluate_performance(data["test"]["label"], y_pred)

```

	precision	recall	f1-score	support
Negative Review	0.51	0.99	0.67	533
Positive Review	0.82	0.03	0.05	533
accuracy			0.51	1066
macro avg	0.66	0.51	0.36	1066
weighted avg	0.66	0.51	0.36	1066

```

# download and install the google-genai library.
pip install -U google-genai

```

```

Requirement already satisfied: google-genai in /usr/local/lib/python3.12/dist-packages (2.12.1)
Requirement already satisfied: anyio<5.0.0,>=4.8.0 in /usr/local/lib/python3.12/dist-packages (from google-genai) (4.14.2)
Requirement already satisfied: google-auth<2.56.0,>=2.48.1 in /usr/local/lib/python3.12/dist-packages (from google-auth[requests]>=2.48.1->google-genai) (2.32.0)
Requirement already satisfied: httpx<1.0.0,>=0.28.1 in /usr/local/lib/python3.12/dist-packages (from google-genai) (0.28.1)
Requirement already satisfied: pydantic<3.0.0,>=2.12.5 in /usr/local/lib/python3.12/dist-packages (from google-genai) (2.13.3)
Requirement already satisfied: requests<3.0.0,>=2.28.1 in /usr/local/lib/python3.12/dist-packages (from google-genai) (2.32.0)
Requirement already satisfied: tenacity<9.2.0,>=8.2.3 in /usr/local/lib/python3.12/dist-packages (from google-genai) (9.1.4)
Requirement already satisfied: websockets<17.0,>=13.0.0 in /usr/local/lib/python3.12/dist-packages (from google-genai) (15.0.7)
Requirement already satisfied: typing-extensions<5.0.0,>=4.14.0 in /usr/local/lib/python3.12/dist-packages (from google-genai) (4.14.0)
Requirement already satisfied: distro<2,>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from google-genai) (1.9.0)
Requirement already satisfied: sniffio in /usr/local/lib/python3.12/dist-packages (from google-genai) (1.3.1)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio<5.0.0,>=4.8.0->google-genai) (3.10.1)
Requirement already satisfied: pyasn1-modules>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from google-auth<2.56.0,>=2.48.1->google-genai) (0.6.1)
Requirement already satisfied: cryptography>=38.0.3 in /usr/local/lib/python3.12/dist-packages (from google-auth<2.56.0,>=2.48.1->google-genai) (45.0.0)
Requirement already satisfied: rsa<5,>=3.1.4 in /usr/local/lib/python3.12/dist-packages (from google-auth<2.56.0,>=2.48.1->google-genai) (4.9.1)

```



```
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1.0.0,>=0.28.1->google-genai)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1.0.0,>=0.28.1->google-genai)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1.0.0,>=0.28.1->google-genai)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.12.2->google-genai)
Requirement already satisfied: pydantic-core==2.46.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.12.2->google-genai)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.12.2->google-genai)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.28.1->google-genai)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.28.1->google-genai)
Requirement already satisfied: cffi>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from cryptography>=38.0.3->google-auth->google-genai)
Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in /usr/local/lib/python3.12/dist-packages (from pyasn1-modules>=0.2.1->google-auth->google-genai)
Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-packages (from cffi>=2.0.0->cryptography>=38.0.3->google-auth->google-genai)
```

```
# # Import the Google GenAI library
from google import genai

# Create client by using api key
client = genai.Client(api_key="AQ.Ab8RN6K0LgYShxrm5CNbkPq8by6Ut0eDHgRXpt_Vxg8jsqnjg")
```

```
# Function to generate a response using the Gemini model
def gemini_generation(prompt, document, model="models/gemini-flash-latest"):
    """Generate an output based on a prompt and an input document."""

    messages = [
        {
            "role": "system",
            "content": "You are a helpful assistant."
        },
        {
            "role": "user",
            "content": prompt.replace("[DOCUMENT]", document)
        }
    ]

    # Gemini expects the conversation as text
    conversation = ""

    # Combine all messages into a single text string
    for message in messages:
        conversation += f"{message['role']}: {message['content']}\n"

    # Send the conversation to the Gemini model
    response = client.models.generate_content(
        model=model,
        contents=conversation    # Input text sent to the model
    )

    # Return only the generated response
    return response.text.strip()
```

```
# Prompt
prompt = """Predict whether the following document is a positive or negative movie review:

[DOCUMENT]

If it is positive return 1 and if it is negative return 0.
Do not give any other answers.
"""

# Predict
# Example movie review to classify
document = "unpretentious, charming, quirky, original"

# # Predict the sentiment using Gemini and print the result
print(gemini_generation(prompt, document))
```

```
1
```

```
# PREDICTIONS ARE NOT POSSIBILE WITH THE ABOVE
```

